



UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Advanced Level

MATHEMATICS

9709/03

Paper 3 Pure Mathematics 3 (P3)

May/June 2007

1 hour 45 minutes

Additional Materials: Answer Booklet/Paper
Graph Paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 75.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of 4 printed pages.



- 1 Expand $(2 + 3x)^{-2}$ in ascending powers of x , up to and including the term in x^2 , simplifying the coefficients. [4]
- 2 The polynomial $x^3 - 2x + a$, where a is a constant, is denoted by $p(x)$. It is given that $(x + 2)$ is a factor of $p(x)$.
- (i) Find the value of a . [2]
- (ii) When a has this value, find the quadratic factor of $p(x)$. [2]
- 3 The equation of a curve is $y = x \sin 2x$, where x is in radians. Find the equation of the tangent to the curve at the point where $x = \frac{1}{4}\pi$. [4]

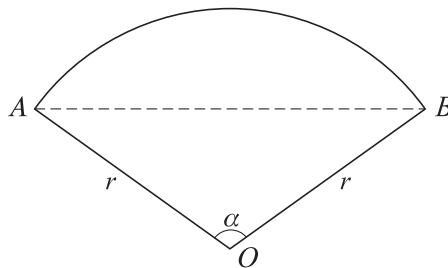
- 4 Using the substitution $u = 3^x$, or otherwise, solve, correct to 3 significant figures, the equation

$$3^x = 2 + 3^{-x}. \quad [6]$$

- 5 (i) Express $\cos \theta + (\sqrt{3}) \sin \theta$ in the form $R \cos(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{1}{2}\pi$, giving the exact values of R and α . [3]

(ii) Hence show that
$$\int_0^{\frac{1}{2}\pi} \frac{1}{(\cos \theta + (\sqrt{3}) \sin \theta)^2} d\theta = \frac{1}{\sqrt{3}}. \quad [4]$$

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The diagram shows a sector AOB of a circle with centre O and radius r . The angle AOB is α radians, where $0 < \alpha < \pi$. The area of triangle AOB is half the area of the sector.

- (i) Show that α satisfies the equation

$$x = 2 \sin x. \quad [2]$$

- (ii) Verify by calculation that α lies between $\frac{1}{2}\pi$ and $\frac{2}{3}\pi$. [2]

- (iii) Show that, if a sequence of values given by the iterative formula

$$x_{n+1} = \frac{1}{3}(x_n + 4 \sin x_n)$$

converges, then it converges to a root of the equation in part (i). [2]

- (iv) Use this iterative formula, with initial value $x_1 = 1.8$, to find α correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

7 Let $I = \int_1^4 \frac{1}{x(4 - \sqrt{x})} dx$.

(i) Use the substitution $u = \sqrt{x}$ to show that $I = \int_1^2 \frac{2}{u(4 - u)} du$. [3]

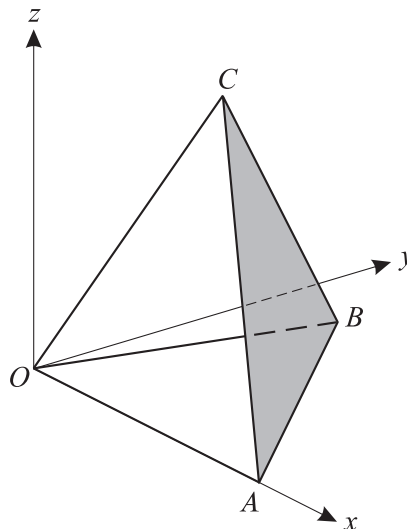
(ii) Hence show that $I = \frac{1}{2} \ln 3$. [6]

8 The complex number $\frac{2}{-1 + i}$ is denoted by u .

(i) Find the modulus and argument of u and u^2 . [6]

(ii) Sketch an Argand diagram showing the points representing the complex numbers u and u^2 . Shade the region whose points represent the complex numbers z which satisfy both the inequalities $|z| < 2$ and $|z - u^2| < |z - u|$. [4]

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The diagram shows a set of rectangular axes Ox , Oy and Oz , and three points A , B and C with position vectors $\vec{OA} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$, $\vec{OB} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ and $\vec{OC} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$.

(i) Find the equation of the plane ABC , giving your answer in the form $ax + by + cz = d$. [6]

(ii) Calculate the acute angle between the planes ABC and OAB . [4]

- 10** A model for the height, h metres, of a certain type of tree at time t years after being planted assumes that, while the tree is growing, the rate of increase in height is proportional to $(9 - h)^{\frac{1}{3}}$. It is given that, when $t = 0$, $h = 1$ and $\frac{dh}{dt} = 0.2$.

(i) Show that h and t satisfy the differential equation

$$\frac{dh}{dt} = 0.1(9 - h)^{\frac{1}{3}}. \quad [2]$$

(ii) Solve this differential equation, and obtain an expression for h in terms of t . [7]

(iii) Find the maximum height of the tree and the time taken to reach this height after planting. [2]

(iv) Calculate the time taken to reach half the maximum height. [1]